

response in JSON format. Please make sure the response is concise and does not include unnecessary explanation.

A safe optimised version could be:

Analyze the customer complaint. Return concise JSON with: *main_issue*, *urgency_level*, *recommended_next_action*.

The optimized prompt removes polite phrasing, repeated conciseness instructions, and wordy transitions. It still retains the task, domain context, output format, and required fields. This is the right balance: the prompt is shorter, but the expected behaviour remains intact.

Adding semantic deduplication

Exact sentence deduplication is useful, but real prompts often repeat ideas in different words. For this reason, a mature framework should support semantic deduplication. One approach is to convert sentences into embeddings and calculate cosine similarity. If two sentences are above a chosen similarity threshold, the optimizer keeps the shorter or clearer sentence. This should be configurable because a threshold that is too low may remove important nuance.

Conceptual flow

Split prompts into sentences, generate embeddings, compare pairwise similarity, remove near-duplicates, rebuild the prompt, and validate that the required intent

markers are still present. In an open source project, this component can be optional so that users may choose lightweight rule-based optimization or embedding-based optimization depending on their accuracy and dependency requirements.

Token monitoring during prompt execution

Token monitoring should be treated as a first-class feature. Before calling the model, the framework can estimate input tokens locally. After the model responds, it can capture actual usage metadata returned by the API. Together, these values help calculate savings, detect abnormal prompts, enforce budget limits, and build dashboards for product teams.

Here's an example of request and response token monitoring:

```
from openai import OpenAI
import tiktoken

client = OpenAI()

def count_tokens(text, model="gpt-4o-mini"):
    encoding = tiktoken.encoding_for_model(model)
    return len(encoding.encode(text))

prompt = "Summarize the incident report in five bullet
```



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